

Test Summary Report

KaloriN AI

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Project Information

Item	Details
Project Name	KaloriN AI
Application Type	Nutrition Tracking Web Application
Testing Type	Manual Testing
Tester	Naufal Fadhiil
Test Management Tool	Qase
Test Environment	Staging / Test Environment
Browsers Tested	Google Chrome, Mozilla Firefox, Microsoft Edge
Execution Date	30 May 2026 - 31 May 2026

1. Executive Summary

KaloriN AI is a web-based nutrition tracking application that provides food analysis, AI-powered food scanning, meal tracking, personalized recommendations, BMI-based nutrition targets, and weekly nutrition insights.

The objective of this testing cycle was to validate the application's functional behavior, business logic, user interface (UI) responsiveness, navigation, profile management, and overall user experience (UX) prior to project completion.

Manual testing was executed across all major user flows and modules. During the testing cycle, several defects regarding data validation, responsiveness, profile management, and error handling were identified and documented within Qase for the development team to resolve.

2. Test Scope

A. In Scope(Tested Modules/Features)

- User & Session Management:
 - Authentication
 - Protected Routes
 - User Onboarding
 - Profile Management

- Core Logic & Features :
 - Home (Dashboard)
 - Analyze
 - Meals
 - Track
 - Insights
- Global System UI/UX:
 - Navigation & Sidebar
 - Global Error Handling
 - Responsiveness

B. Out of Scope

The following testing activities were excluded from this specific testing cycle:

- Security & Penetration Testing
- Performance, Load, & Stress Testing

3. Test Execution Summary

A. Execution Statistics

Metric	Result	Percentage
Total Test Cases	122	100%
Executed	122	100%
Passed	110	90.16%
Failed	10	8.20%
Skipped	2	1.64%
Blocked	0	0%

B. Execution Result Overview

- The majority of functional test cases passed successfully. Core product features such as user authentication, routing protection, the onboarding flow, food analysis, meal recommendations, and calorie tracking performed according to specifications.
- Failed test cases were predominantly concentrated within the Profile Management module, input validation rules, user interface responsiveness on smaller viewports, and global error handling mechanisms.

4. Defect Summary

A total of **10 defects** were discovered and documented during the testing activities.

A. Defect Distribution by Severity

-  **Critical (3)**: Defects that break core business logic or trigger severe system failures.
-  **Major (4)**: Significant functional impairment of primary features without causing system crashes.
-  **Minor (2)**: Minor UI/UX discrepancies that do not obstruct the primary user transaction flows.
-  **Trivial (1)**: Cosmetic issues or visual refinements that do not impact functionality.

B. Identified Defects List

Bug ID	Title	Severity	Module / Area / Feature
BUG-001	Username inconsistent after registration	Major	Profile Management
BUG-002	Navbar not displayed in Chrome browser	Major	Navigation & Sidebar
BUG-003	Scan & Log Meal button overlaps profile section	Minor	Responsiveness

BUG-004	Male user can select pregnant and breastfeeding options	Major	Business Logic / Profile
BUG-005	Weight input accepts unrealistic values	Critical	Data Validation / Profile
BUG-006	Height input accepts unrealistic values	Major	Data Validation / Profile
BUG-007	Future birthdate generates negative age	Critical	Data Validation / Onboarding / Profile
BUG-008	Profile image not displayed consistently in sidebar	Minor	State Management / Firebase
BUG-009	No proper error handling when internet connection is lost	Critical	Global Error Handling
BUG-010	Default browser validation popup used instead of custom validation UI	Trivial	UI Polish / Form

5. Key Findings

A. Data Validation & Boundary Issues

- The weight field accepts unrealistic values.
- The height field accepts unrealistic values.
- Invalid future birthdates can be submitted.
- Negative age calculation occurs after saving future birthdates.

B. Business Logic Anomalies

Gender specific rule failure "Male" as their gender to toggle "pregnant" and "breastfeeding" status options. This inconsistency threatens the integrity of the AI-driven calorie and meal recommendation calculations.

C. UI & Responsiveness Issues

- Sidebar components overlap content on smaller viewports.
- Profile image is not displayed consistently across all application sections.

D. Browser Compatibility & Network Resilience

- The application displays a blank page when internet connectivity is lost.
- The application lacks dedicated offline state handling. Losing internet connectivity while reloading a page throws a raw white blank page.

6. Risk Assessment

Module	Risk Level	Justification
Profile Management	High	Contains multiple high-severity validation and business logic defects that directly disrupt user profile accuracy and BMI calculations.
Error Handling	Medium	Displaying blank screens during network drops can severely damage user retention and trust.
User Onboarding	Medium	The future birthdate bug occurs at the very first stage of the user interaction lifecycle.
Navigation & Sidebar	Medium	Rendering issues on Google Chrome degrade basic navigation usability across primary screens.

Authentication	Low	Functions securely and reliably; protected route mechanisms are 100% effective.
Core Features (AI Scanner)	Medium	Additionally, the AI Food Scanner feature presents a medium risk due to observed inconsistencies in recognizing certain Indonesian food categories. Model accuracy may be affected by dataset coverage and training data quality.

7. Quality Assessment

A. Strengths

- The registration, login, and protected route infrastructures are exceptionally stable and secure.
- The primary food database query and AI scanning features successfully identify and process nutritional logs as intended.
- The meal recommendation system and weekly insight graph rendering are highly accurate, provided the initial profile boundary data is valid.

B. Areas Requiring Improvement

- Enforcement of strict upper and lower boundaries for numeric user inputs (Input Boundary Validation).
- Enforcement of conditional UI business rules (e.g., gender-to-medical-status relationships).
- Graceful degradation of user experience during sudden network connectivity loss.
- Layout consistency and element layer control across fluid device viewports.
- AI food recognition accuracy for local Indonesian foods.
- Dataset coverage and training data diversity.

8. AI Model Observation

During exploratory testing of the AI Food Scanner feature, several inconsistencies were observed when scanning local Indonesian foods. Examples included foods such as:

- Nasi Goreng
- Rendang
- And another Indonesian traditional meals

Based on testing observations, this issue appears to be related to dataset coverage and training data quality rather than application functionality or system integration.

9. Conclusion

Manual testing was successfully completed for all planned modules within the project scope. Testing execution achieved a completion rate of 100%, with 110 test cases passing and 10 test cases failing.

The application's primary business functionality is operational and suitable for academic demonstration, capstone presentation, and further development activities.

However, several defects related to profile validation, business rules, responsiveness, browser compatibility, and error handling should be addressed before any production deployment or public release.

10. Recommendation

1. Implement validation limits for weight and height inputs.
2. Prevent future birthdate submissions.
3. Enforce gender-based validation rules for pregnancy and breastfeeding options.
4. Improve responsive sidebar behavior on smaller screens.
5. Fix profile image synchronization across application pages.
6. Implement custom validation messages aligned with application design.
7. Add dedicated offline and network error handling pages.
8. Perform regression testing after defect fixes are deployed.
9. Improve AI food recognition accuracy by expanding the training dataset with more Indonesian food categories and image variations.
10. Conduct periodic model evaluation and retraining using representative local food datasets.

Final Testing Status

Testing Cycle Status:	COMPLETED
Go/No-Go Recommendation:	GO (With Conditions)
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